Calculation questions – 4.4 Atomic structure

AQA GCSE Physics (8463) Higher · Paper 1 · 12 questions · 31 marks · answers at the back

Equations for this topic

Equation	Symbols and units	Exam
After n half-lives: amount left = start ÷ 2^n		Rule
Alpha decay: mass number − 4, atomic number − 2		Rule
Beta decay: mass number unchanged, atomic number + 1		Rule
Gamma: no change to either number		Rule
Corrected count = measured count − background count		Rule

Write the equation, substitute, then rearrange. Convert to base units first. Give a unit and 2–3 significant figures. Take g = 9.8 N/kg.

Grade 4–5

1 An isotope has a half-life of 6 hours. What fraction of it remains after 18 hours? [2 marks]

Answer: _____ Unit: _____

2 A sample has an activity of 800 Bq and a half-life of 2 days. Calculate its activity after 6 days. [2 marks]

Answer: _____ Unit: _____

3 A nucleus contains 92 protons and 146 neutrons. State its mass number and atomic number. [2 marks]

Answer: _____ Unit: _____

Grade 6–7

4 The activity of a source falls from 640 Bq to 40 Bq in 20 minutes. Calculate its half-life. [3 marks]

Answer: _____ Unit: _____

5 Uranium-238 (atomic number 92) decays by alpha emission to thorium. Give the mass number and atomic number of the thorium nucleus. [2 marks]

Answer: _____ Unit: _____

6 Carbon-14 (atomic number 6) decays by beta emission to nitrogen. Give the mass number and atomic number of the nitrogen nucleus. [2 marks]

Answer: _____ Unit: _____

7 A 2.4 g sample has a half-life of 8 days. Calculate the mass of the isotope left after 32 days. [2 marks]

Answer: _____ Unit: _____

8 A Geiger counter reads 510 counts/min near a source. The background count is 30 counts/min. The source's half-life is 40 minutes. Calculate the reading on the counter after 2 hours. [4 marks]

Answer: _____ Unit: _____

Grade 8–9

9 Calculate the percentage of a radioactive isotope that (a) remains and (b) has decayed after 5 half-lives. [3 marks]

Answer: _____ Unit: _____

10 A rock contains 1 part potassium-40 to every 3 parts of its decay product argon-40. All the argon came from decay of potassium. The half-life of potassium-40 is 1.3 billion years. Calculate the age of the rock. [3 marks]

Answer: _____ Unit: _____

11 A medical tracer has an activity of 1200 Bq and a half-life of 15 hours. It is safe once the activity is below 100 Bq. Calculate the minimum number of whole half-lives, and the time in hours, before it is safe. [3 marks]

Answer: _____ Unit: _____

12 Radium-226 (atomic number 88) undergoes two alpha decays and then one beta decay. Calculate the mass number and atomic number of the final nucleus. [3 marks]

Answer: _____ Unit: _____

Answers – 4.4 Atomic structure

Mark yourself honestly: a mark for the right equation, a mark for substitution, a mark for the answer with its unit.

Q	Working	Answer
1	18 ÷ 6 = 3 half-lives; $(½)^3$	**1/8**
2	3 half-lives: 800 → 400 → 200 → 100	**100 Bq**
3	92 + 146	**mass number 238, atomic number 92**
4	640 → 320 → 160 → 80 → 40 = 4 half-lives in 20 min	**5.0 min**
5	alpha: mass − 4, atomic − 2	**Th-234, atomic number 90**
6	beta: mass same, atomic + 1	**N-14, atomic number 7**
7	4 half-lives; 2.4 ÷ 2^4	**0.15 g**
8	corrected = 480; 120 min = 3 half-lives → 60; add background 30	**90 counts/min**
9	$(½)^5$ = 1/32 = 0.03125	**(a) 3.1% (b) 96.9%**
10	started with 4 parts, 1 left = ¼ → 2 half-lives; 2 × 1.3	**2.6 billion years**
11	1200 → 600 → 300 → 150 → 75: 4 half-lives; 4 × 15	**4 half-lives, 60 hours**
12	mass 226 − 8 = 218; atomic 88 − 4 + 1 = 85	**mass 218, atomic number 85**

Common mistakes. Halving the background count (Q8: subtract it first, add it back last); reading "1 to 3" as one-third left (Q10: it is one-quarter).